Cover Letter

To: The Editor-in-Chief
Journal: *Remote Sensing* (MDPI)
Date: October 18, 2026

Dear Editor,

We are pleased to submit our manuscript entitled "When F1 and ROC-AUC Disagree: A 30-Point Metric Divergence Under Spatial Interleaving in Extreme-Imbalance Bi-Temporal Change Detection" for consideration for publication in *Remote Sensing*.

This paper documents a spatial-evaluation failure mode in bi-temporal change detection, measured in a single training region (Vietnam; $\sim$60 eligible patches): a spatially interleaved random split produces a 29.6-percentage-point higher change $F1$ than a buffer-separated test assignment, positive in all ten partitions, while the corresponding ROC-AUC difference is only $+1.9$ pp. The audit quantifies this in a Sentinel-2 mangrove change-detection setting (0.8–2.6% change prevalence), decomposes the gap into a small prevalence-arithmetic component ($+2.7$ pp) and a dominant recall shift associated with the joint spatial configuration of training and test samples, and states the identification boundary of what the design can and cannot attribute.

Three contributions, corresponding to the abstract, are the quantified $F1$–ROC-AUC divergence under extreme imbalance; the 2×2 train/test factorial bounding the prevalence-arithmetic contribution at $\sim 9\%$ with the training$\times$test interaction exceeding either main effect; and the demonstration that nine of 90 runs score near-zero $F1$ yet rank at ROC-AUC 0.77–0.97, so a near-zero $F1$ under extreme imbalance does not imply an uninformative model.

Three scope limits are disclosed in the manuscript and we ask referees to read the claims within them: (i) a single training region, so the effective number of independent spatial units is small (effective n=2–5 in the controls) and the magnitudes are region-specific; (ii) a single GMW reference label, so all scores are agreement with a noisy reference rather than field accuracy, with source-to-source difference larger than the signal; (iii) the checkpoint-selection channel of the interleaved arm (early stopping on a validation set proximate to its test set) is a component of the interleaved protocol by construction, and the fixed-epoch control that would bound it is deferred to a follow-up, as stated in the Limitations section.

One further item is part of the submitted package: the square-root-weighting replication (150 result files) is listed in the experiment inventory of the accompanying repository.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have read and approved the submitted version. The full code base and results are deposited at `https://github.com/hao-feier/split-audit`.

Thank you for your consideration.

Sincerely

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